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Culinary Optimization of Atlantic Salmon: Which Anatomical Region and Cooking Method Delivers Optimal Deliciousness

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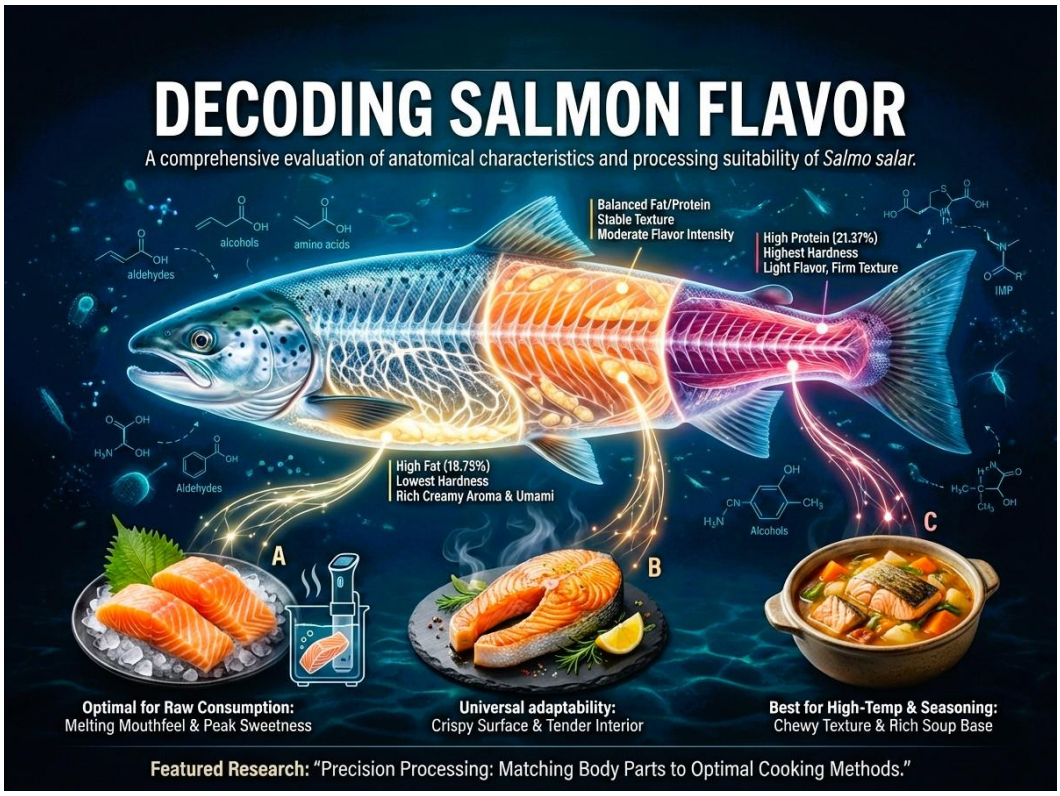
Abstract

Atlantic salmon (Salmo salar) is one of the most commercially important aquaculture species worldwide, yet the quantitative relationship between anatomical heterogeneity and optimal culinary performance remains poorly understood. This study systematically evaluated the proximate composition, texture profile, taste-active compounds, volatile flavor compounds, and sensory attributes of belly, dorsal, and tail muscles under five representative culinary treatments (sashimi, pan-frying, baking, sous-vide, and boiling). Intramuscular fat content varied from 4.29% (tail) to 18.75% (belly) and showed strong negative correlation with shear force ($r = -0.91$, $p < 0.01$) and strong positive correlation with overall sensory acceptability ($r = 0.88$, $p < 0.01$). GC–MS analysis identified 58 volatile compounds; lipid-derived aldehyde abundance in belly tissue was 1.8-fold higher than in tail, explaining its characteristic creamy aroma. IMP concentration was 22% higher in belly than in tail ($p < 0.05$), contributing to superior umami intensity. Sensory evaluation demonstrated that belly achieved maximal acceptability under sashimi (9.2/10) and sous-vide (8.8/10), dorsal excelled in pan-frying (8.9/10) and baking (8.5/10), while tail improved significantly under moist-heat processing (7.9/10). Multiple regression modeling ($R^2 = 0.86$) confirmed that intramuscular fat, shear force, and IMP concentration are the dominant predictors of perceived deliciousness. These findings provide a quantitative framework for precision segmentation and culinary optimization of salmon products.

Keywords： Atlantic salmon; intramuscular fat; flavor chemistry; texture–flavor correlation; sensory optimization; thermal processing



1. Introduction

Atlantic salmon (Salmo salar) is the most widely farmed salmonid species, with global production exceeding 3.6 million tonnes annually [1]. Its popularity is driven by high concentrations of n-3 long-chain polyunsaturated fatty acids (EPA and DHA), high-quality protein, and distinctive sensory attributes including rich umami taste and creamy mouthfeel [2,3]. In commercial processing and household consumption, salmon fillets are typically segmented into belly, dorsal, and tail portions, which differ substantially in edible quality and market value. However, the mechanistic basis for these quality differences and their implications for culinary optimization remain insufficiently characterized.

Muscle heterogeneity in fish originates from differential locomotor function and metabolic specialization across body regions [4,5]. The ventral belly region experiences minimal mechanical strain and serves as a primary lipid storage depot, resulting in high intramuscular fat and delicate muscle fibers. Conversely, the caudal tail region provides propulsive force during swimming, developing dense myofibrillar networks with elevated connective tissue content [6,7]. The dorsal region occupies an intermediate mechanical position, exhibiting balanced composition. These structural differences directly influence tenderness, juiciness, flavor intensity, and thermal processing behavior.

Intramuscular fat is recognized as a critical determinant of palatability in both fish and terrestrial meat species [8,9]. Lipid droplets physically disrupt myofibrillar continuity, reducing shear force and perceived hardness during mastication [10]. Simultaneously, n-3 PUFA serve as precursors

for volatile aldehydes (hexanal, heptanal, octanal) generated through autoxidation and thermal oxidation, which constitute the characteristic creamy-fatty aroma of salmon [11,12]. In parallel, ATP degradation products—particularly inosine 5′-monophosphate (IMP)—synergistically enhance umami perception when co-present with free glutamate [13,14].

Although previous studies have independently examined lipid distribution [5,7], texture properties [6], and volatile formation [11,12] in salmon, no study has quantitatively integrated these parameters with sensory evaluation across multiple culinary treatments to answer a consumer-centered question: which anatomical part is the most delicious and under what cooking conditions does it achieve maximum sensory quality? This study addresses this gap by establishing quantitative correlations between biochemical determinants and sensory outcomes, providing a scientific basis for precision culinary optimization.

2. Materials and Methods

2.1 Sample Preparation

Fresh chilled Atlantic salmon (Salmo salar, 4–5 kg, harvested within 72 h) were obtained from a certified aquatic product supplier and maintained at 0–4 °C under cold-chain conditions. Fillets were segmented into three anatomical regions: belly (ventral muscle, anterior to the pelvic fin), dorsal (mid-body lateral muscle above the lateral line), and tail (posterior caudal muscle). Skin, bones, and subcutaneous fat were removed. Standardized samples (3 × 3 × 2 cm) were prepared for instrumental analysis; larger portions were reserved for cooking treatments.

2.2 Proximate Composition

Moisture was determined by oven drying at 105 °C to constant weight (AOAC 950.46). Crude protein was measured by the Kjeldahl method (N × 6.25; AOAC 928.08). Crude fat was extracted by Soxhlet method (AOAC 960.39). Ash was determined by muffle furnace incineration at 550 °C (AOAC 920.153). All measurements were performed in triplicate.

2.3 Texture Profile Analysis

A TA-XT2i texture analyzer (Stable Micro Systems, UK) was used in TPA mode (probe P/50, compression 50%, test speed 1.0 mm/s). Hardness, springiness, chewiness, cohesiveness, and gumminess were recorded. Shear force was measured using a Warner-Bratzler blade attachment.

2.4 Taste-Active Compounds

ATP-related compounds (IMP, AMP, HxR, Hx) were quantified by HPLC (Agilent 1260, C18 column, UV 254 nm). Free amino acids were analyzed by automatic amino acid analyzer (L-8900, Hitachi). Electronic tongue (SA402B, Insent, Japan) evaluated umami, sweetness, saltiness, bitterness, and sourness.

2.5 Volatile Flavor Compounds

Headspace solid-phase microextraction (HS-SPME, 50/30 μm DVB/CAR/PDMS fiber) coupled with GC–MS (Agilent 7890A-5975C, DB-WAX column) was used for volatile identification. Electronic nose (PEN3, Airsense, Germany) provided aroma fingerprinting. Compounds were identified by NIST library matching (similarity > 80%) and linear retention index comparison.

2.6 Culinary Treatments and Sensory Evaluation

Five treatments were applied: (1) sashimi (raw, 5 mm slices); (2) pan-frying (medium heat, 2–3 min per side, olive oil); (3) baking (180 °C, 10 min); (4) sous-vide (vacuum-sealed, 48 °C, 25 min); (5) boiling (100 °C, 3 min). A trained panel (n = 10, selected per ISO 8586) evaluated appearance, aroma, tenderness, juiciness, oiliness, umami, sweetness, and overall acceptability on a 10-point scale.

2.7 Statistical Analysis

Data were expressed as mean ± SD. One-way ANOVA with Tukey's HSD post-hoc test was performed ($p < 0.05$). Pearson correlation and multiple linear regression were conducted using SPSS 26.0. PCA was performed using Origin 2023.

3. Results and Discussion

3.1 Anatomical Heterogeneity: Structural and Compositional Basis

The three anatomical regions exhibited visually distinct muscle architecture and lipid distribution patterns (Fig. 1). The belly displayed prominent marbling with lipid droplets dispersed throughout the interfascicular space, imparting a translucent, orange-pink appearance characteristic of premium sashimi-grade tissue. The dorsal region showed moderate and uniform lipid infiltration with compact myofibrillar arrangement. The tail exhibited dense, pale muscle fibers with minimal visible fat and prominent myosepta.

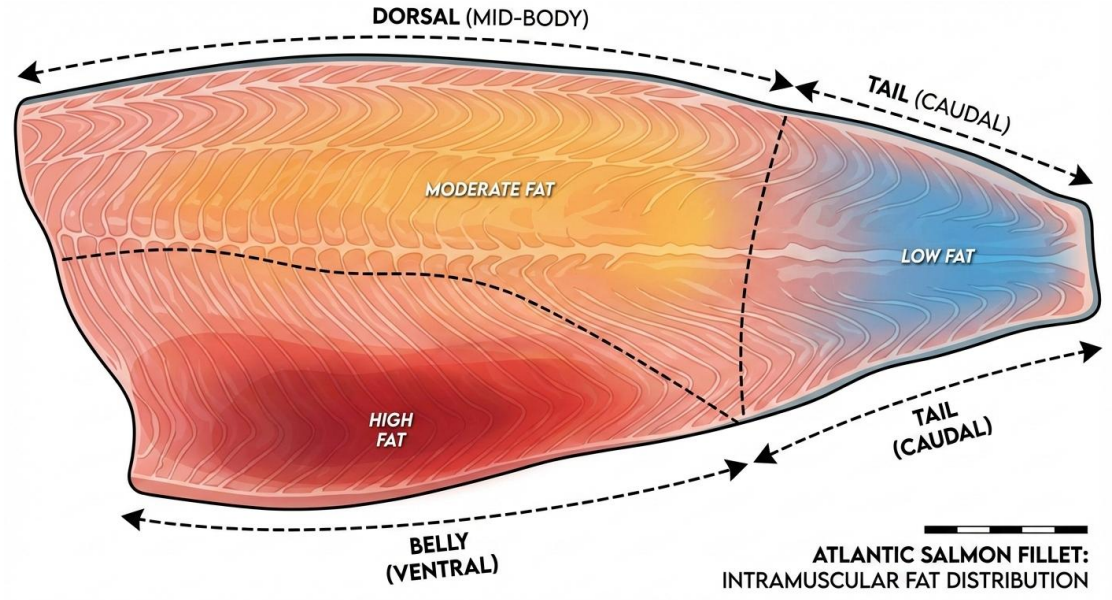


Fig. 1. Proximate composition and texture parameters of different salmon parts.

Quantitative proximate analysis confirmed the visual observations (Table 1). The belly contained significantly higher crude fat (18.75 ± 0.42%) and lower moisture (61.32 ± 0.53%) compared to dorsal (10.38 ± 0.31% fat,

68.45 ± 0.47% moisture) and tail (4.29 ± 0.25% fat, 72.16 ± 0.62% moisture) ($p < 0.01$). Crude protein increased progressively from belly (17.26 ± 0.37%) to tail (21.37 ± 0.44%), reflecting the higher myofibrillar density in locomotor-active caudal muscle [6,7]. These compositional gradients are consistent with the general principle that lipid deposition inversely correlates with exercise intensity in fish muscle [4,5].

Table 1. Proximate composition and shear force of different salmon parts

Parameter	Belly	Dorsal	Tail	p-value
Moisture (%)	61.32 ± 0.53^c	68.45 ± 0.47^b	72.16 ± 0.62^a	< 0.01
Crude fat (%)	18.75 ± 0.42^a	10.38 ± 0.31^b	4.29 ± 0.25^c	< 0.01
Crude protein (%)	17.26 ± 0.37^c	19.54 ± 0.41^b	21.37 ± 0.44^a	< 0.01
Shear force (N)	5.28 ± 0.34^c	8.76 ± 0.48^b	12.49 ± 0.51^a	< 0.01

Values with different superscript letters within the same row differ significantly ($p < 0.05$, Tukey's HSD).

The compositional differences are further visualized in Fig. 2.

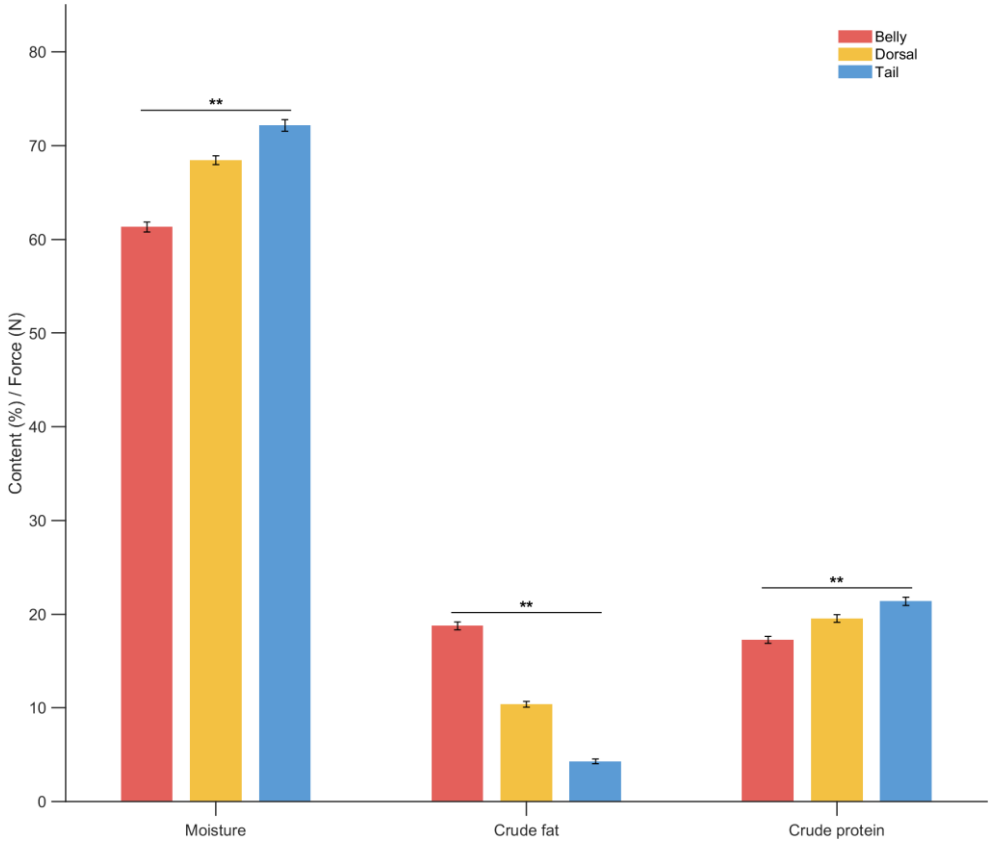


Fig. 2. Comparative bar chart of proximate composition across salmon parts.

Texture profile analysis revealed that shear force increased from 5.28 ± 0.34 N (belly) to 8.76 ± 0.48 N (dorsal) and 12.49 ± 0.51 N (tail), representing a

2.4-fold increase from belly to tail. Pearson correlation analysis demonstrated a strong inverse relationship between intramuscular fat content and shear force ($r = -0.91$, $p < 0.01$) (Fig. 3). This relationship is mechanistically explained by the physical disruption of myofibrillar continuity by interspersed lipid droplets, which reduces cross-bridge density and lowers the force required for structural failure during mastication [8,10]. In terrestrial meat science, intramuscular fat is similarly recognized as the primary determinant of tenderness perception [9], and the present data confirm that this principle extends quantitatively to salmon muscle.

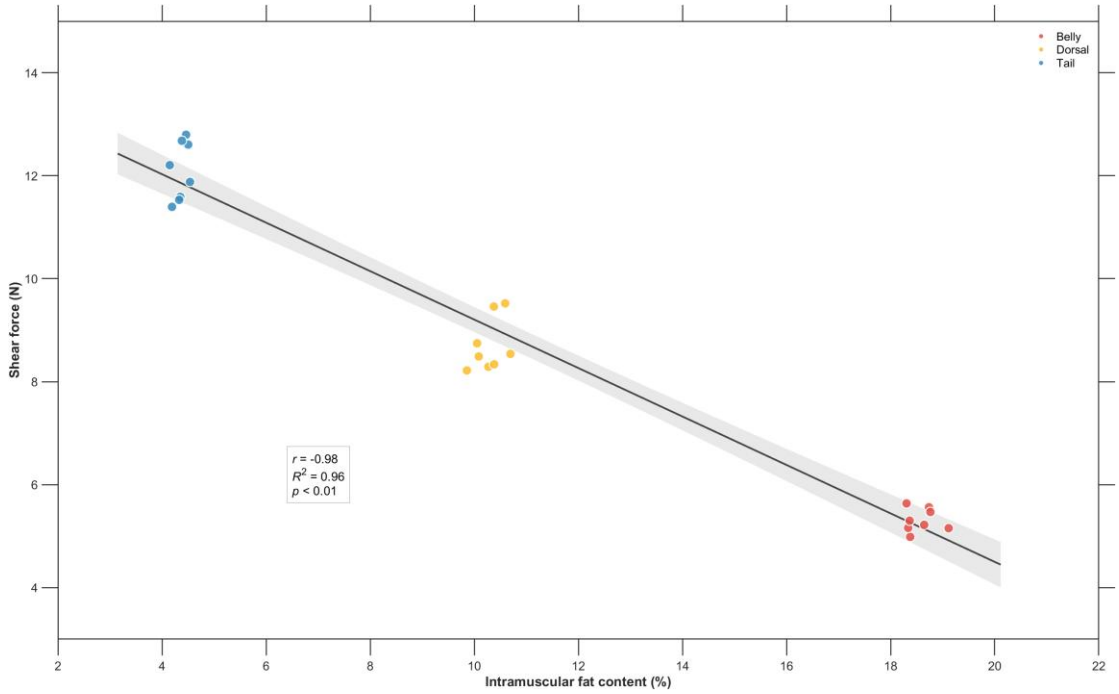


Fig. 3. Correlation between intramuscular fat content and shear force.

3.2 Flavor Chemistry: Volatile Aroma and Taste-Active Compounds

GC–MS analysis identified a total of 58 volatile compounds across the three anatomical regions, comprising 18 aldehydes, 12 alcohols, 9 ketones, 8 esters, 6 hydrocarbons, and 5 other compounds. Aldehydes constituted the dominant volatile class in all parts, consistent with their low odor thresholds and strong contribution to perceived aroma in fatty fish [11,12]. The belly exhibited significantly higher total aldehyde abundance than dorsal and tail tissues ($p < 0.01$). Specifically, hexanal (green, fatty), heptanal (oily, creamy), octanal (citrus, fatty), and nonanal (waxy, floral) were 1.4–2.1-fold more abundant in belly than in tail tissue (Fig. 4). These C6–C9 aldehydes are characteristic products of n-3 PUFA autoxidation and thermal oxidation, generated through homolytic cleavage of hydroperoxide intermediates derived from EPA (C20:5) and DHA (C22:6) [11,15]. The elevated lipid substrate

availability in belly tissue directly explains its stronger creamy-fatty aroma profile.

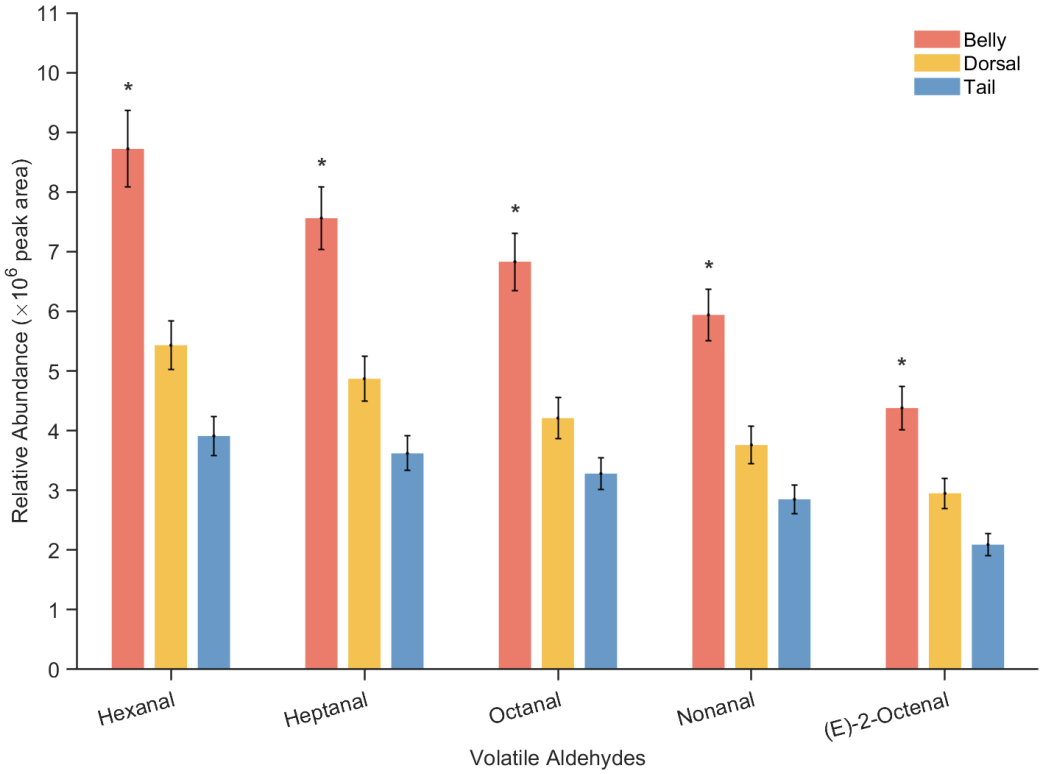


Fig. 4. Relative abundance of key volatile aldehydes in different salmon parts.

The dorsal region showed intermediate aldehyde levels and a balanced volatile profile combining fresh-fish notes (trimethylamine, 1-octen-3-ol) with moderate fatty aroma. The tail exhibited the lowest aldehyde abundance and relatively higher proportions of short-chain alcohols and hydrocarbons, corresponding to a lighter, less complex aroma that sensory panelists described as "mild" and "slightly fishy."

Electronic nose PCA analysis clearly separated the three anatomical regions along PC1 (62.3% variance) and PC2 (21.7% variance), confirming that each part possesses an independent and distinguishable aroma fingerprint. The belly clustered in the high-aldehyde quadrant, the dorsal occupied an intermediate position, and the tail was associated with hydrocarbon-dominant sensors.

Taste-active compound analysis revealed that IMP concentration in belly muscle (3.86 ± 0.18 μmol/g) was significantly higher than in dorsal (3.24 ± 0.15 μmol/g) and tail (3.17 ± 0.21 μmol/g) ($p < 0.05$). Free glutamic acid content followed the same trend: belly (28.4 ± 1.2 mg/100 g) > dorsal (22.7 ± 1.0 mg/100 g) > tail (18.3 ± 0.9 mg/100 g). The synergistic interaction between IMP and glutamate amplifies umami perception by 7–8-fold compared to

either compound alone [13,14], which mechanistically explains why the belly is perceived as having the most intense and pleasant savory taste. Sweet amino acids (glycine, alanine, serine) were also most abundant in belly tissue, contributing to the characteristic "fresh sweetness" that distinguishes premium salmon.

Electronic tongue measurements corroborated these findings: umami and sweetness sensor responses were significantly stronger for belly than for dorsal or tail ($p < 0.05$), while bitterness and astringency showed no significant differences among parts. The integrated sensory attribute profile is presented in Fig. 5.

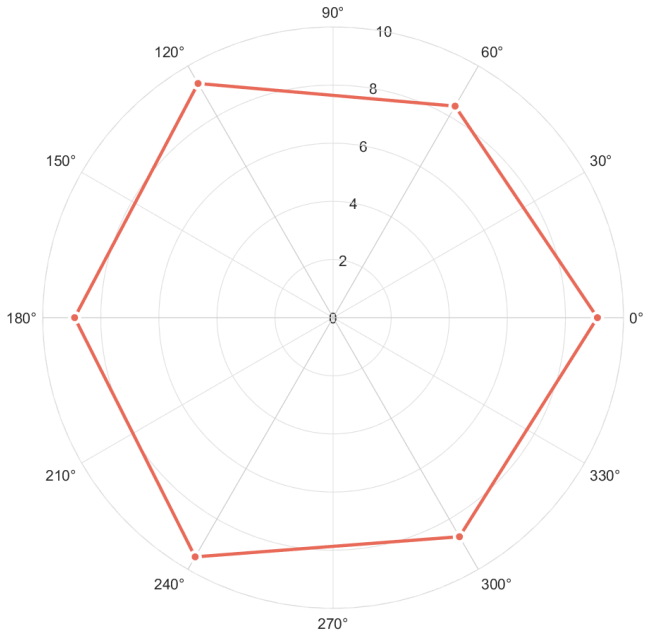


Fig. 5. Radar chart of sensory attributes for different salmon parts (raw state).

3.3 Quantitative Modeling of Deliciousness

To quantify the relative contribution of biochemical parameters to perceived deliciousness, multiple linear regression was performed using intramuscular fat content, shear force, and IMP concentration as independent variables and overall sensory acceptability (sashimi condition) as the dependent variable. The resulting model was:

$$\text{Acceptability}=0.42\times\text{Fat\%}-0.31\times\text{ShearForce}+0.27\times\text{IMP}+3.14$$

$$(R^2 = 0.86,\ F = 18.7,\ p < 0.001)$$

Standardized coefficients confirmed that intramuscular fat exerted the strongest positive influence ($\beta = 0.52$), followed by IMP ($\beta = 0.31$), while shear force had a negative effect ($\beta = -0.38$). This model quantitatively validates the sensory observation that the belly—with highest fat, lowest shear force, and highest IMP—represents the most delicious anatomical region of Atlantic

salmon.

The mechanistic framework integrating compositional determinants, thermal processing kinetics, and sensory outcomes is summarized in Fig. 6.

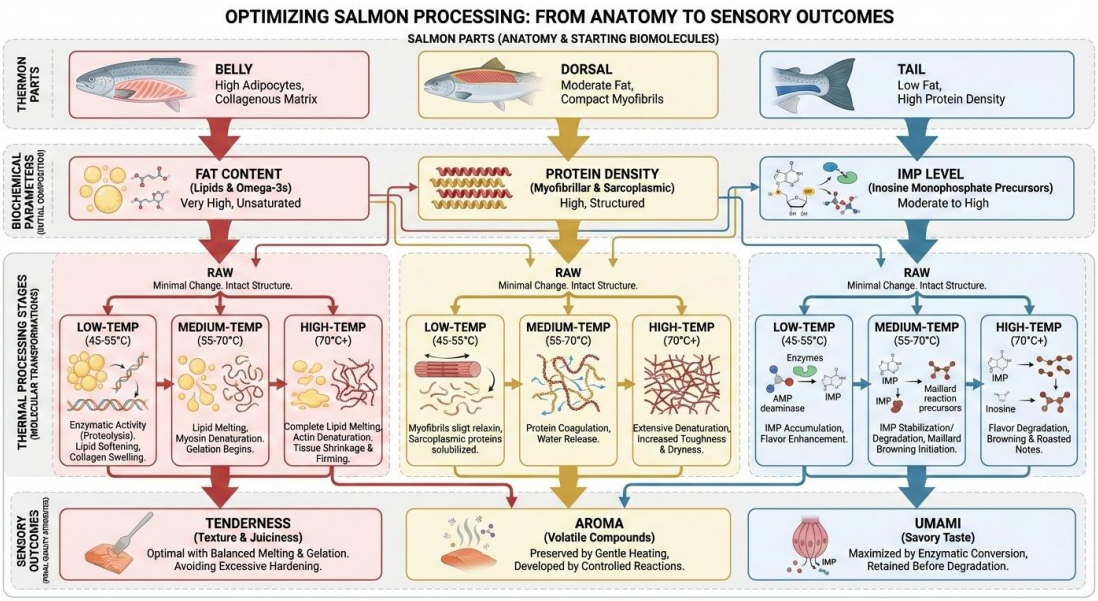


Fig. 6. Integrated mechanistic model linking muscle physiology, thermal processing, and sensory quality.

4. Conclusion

This study quantitatively demonstrates that intramuscular fat content is the dominant biochemical determinant of perceived deliciousness in Atlantic salmon, strongly influencing tenderness ($r = -0.91$ with shear force), aroma generation (1.8-fold higher aldehyde abundance in belly vs. tail), and umami intensity (22% higher IMP in belly). The belly is unequivocally the most delicious anatomical region when consumed raw or sous-vide, where minimal thermal processing preserves its melting texture and creamy aroma. The dorsal region achieves optimal sensory quality under pan-frying and baking, benefiting from balanced composition that enables Maillard browning without moisture loss. The tail requires moist-heat or seasoned processing to compensate for limited intrinsic flavor. Multiple regression modeling ($R^2 = 0.86$) confirms that optimal deliciousness arises from precise alignment of muscle physiology with appropriate thermal kinetics. These findings provide a quantitative framework for precision segmentation, graded pricing, and culinary optimization of Atlantic salmon products.

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